

The Romer Model

Note: This content will help you once you have gone through the book. If you did not read this model from the book before, you may face some trouble. In that case, check that particular topic from the book or simply ask someone to explain the issue who have the habit of reading books. Also do not use any unfair means in the exam using this content (or any content).

[This note is due to an extraordinary hard work by Tushar Siddik and Vaskor Dey Dhrubo. If you find any mistake, please notify them]

The Romer Model

Before going through this note, think of an idea. An idea that helps manufacturing a product.

For example, the idea that “the force produced by steam pressure can be used to push a piston back and forth in a cylinder” is central to the development of steam engine. This idea can be seen as an input in the production of steam engine.

Therefore, other than labor and capital, there may be another type of input that embodies an idea. In the following text you will find terms like “input embodying an idea”, “input embodying different ideas” etc. Now you know what that means.

3.5 The Romer Model

Overview

- The model of **P. Romer (1990)** of **endogenous technological change**
- In this model, R&D is under taken by **profit-maximizing economic factors**
- R&D fuels growth
 - which in turn affects the incentives for devoting resources to R&D
- Any model where the creation of knowledge is motivated by the returns
 - the knowledge commands in the market must involve **departures from perfect competition**
- If knowledge is sold at marginal cost
 - the creators of knowledge earn negative profits
- Romer deals with this issue by assuming that
 - knowledge consists of distinct ideas and
 - Inputs into production that embody different ideas are imperfect substitutes.
 - He also assumes that the developer of an idea has monopoly rights to the use of the idea.
- These assumptions imply that
 - the developer can charge a price above marginal cost for the use of his or her idea.
 - The resulting profits provide the incentives for R&D.
- The reason for constructing the model this way is
 - it simplifies the analysis dramatically
 - Models of this type exhibit **no transition dynamics**.
 - In response to a shock, the economy jumps immediately to its new BGP.
- Two types of simplifications of the model
 - The **first** are assumptions about **functional forms and parameter values**
 - The model in P. Romer (1986) that launched new growth theory is closely related to our **learning-by-doing model with $\phi = 1$ and $n = 0$** (analogous to the assumptions of $\theta = 1$ and $n = 0$ in R&D Model without capital)
 - The other is the **elimination of all types of physical and human capital**

The Ethier Production Function and the Returns to Knowledge Creation

- The first step in presenting the model is to describe
 - how knowledge creators have market power.
- Thus for the moment
 - we take the **level of knowledge** as **given** and
 - describe how **inputs embodying different ideas combine to produce final output**.
- We assume that there is a **range of ideas** that are currently available that extends from **0 to A**, where $A > 0$
- We will use $L(i)$ to denote both
 - the **quantity of labor** devoted to **producing input i** and
 - the **quantity of input i** that **goes into final-goods production**.

For ideas that have not yet been discovered (that is, for $i > A$), inputs embodying the ideas cannot be produced at any cost.

The specific assumption about how the inputs combine to produce final output uses the **production function proposed by Ethier (1982)**:

$$Y = \left[\int_{i=0}^A L(i)^\varphi di \right]^{\frac{1}{\varphi}} \quad 0 < \varphi < 1 \quad (3.28)$$

Where, φ = substitutability among inputs

To see the implications of this function,

- let L_Y denote the **total number of workers producing inputs**, and
- suppose the number producing each available input is the same.
- Then $L(i) = \frac{L_Y}{A}$ for all i , and so

$$\begin{aligned} Y &= \left[A \left(\frac{L_Y}{A} \right)^\varphi \right]^{\frac{1}{\varphi}} \\ &= A^{\frac{1-\varphi}{\varphi}} L_Y \end{aligned} \quad (3.29)$$

Explanation:

$$Y = [\int_{i=0}^A L(i)^\varphi di]^\frac{1}{\varphi}$$

$$Y = [\int_{i=0}^A (\frac{L_Y}{A})^\varphi di]^\frac{1}{\varphi}$$

$$Y = [(\frac{L_Y}{A})^\varphi \int_{i=0}^A di]^\frac{1}{\varphi}$$

$$Y = [[(\frac{L_Y}{A})^\varphi \cdot i]_0^A]^\frac{1}{\varphi}$$

$$Y = [(\frac{L_Y}{A})^\varphi \cdot A - (\frac{L_Y}{A})^\varphi \cdot 0]^\frac{1}{\varphi}$$

$$Y = [A (\frac{L_Y}{A})^\varphi]^\frac{1}{\varphi}$$

$$Y = [A \frac{L_Y^\varphi}{A^\varphi}]^\frac{1}{\varphi}$$

$$Y = [A^{(1-\varphi)} L_Y^\varphi]^\frac{1}{\varphi}$$

$$Y = A^{\frac{1-\varphi}{\varphi}} L_Y^\frac{\varphi}{\varphi}$$

$$Y = A^{\frac{1-\varphi}{\varphi}} L_Y$$

- This expression has **two critical implications**.
 - First, there are **constant** returns to L_Y
 - Second, output is **increasing** in A
- Consider **the cost-minimization problem** of a representative output producer.
- Let $p(i)$ denote **the price charged**
 - by the holder of the patent on idea i
 - for each unit of the input embodying that idea.

The Lagrangian for the problem of producing one unit of output at minimum cost is

$$\mathcal{L} = \int_{i=0}^A p(i)L(i)di - \lambda \{[\int_{i=0}^A L(i)^\varphi di]^\frac{1}{\varphi} - 1\} \quad (3.30)$$

Explanation:

Minimize: $\int_{i=0}^A p(i)L(i)di$

Subject to: $[\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}} = 1$

Therefore, the lagrangian is:

$$\mathcal{L} = \int_{i=0}^A p(i)L(i)di + \lambda \{1 - [\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}}\}$$

$$\mathcal{L} = \int_{i=0}^A p(i)L(i)di - \lambda \{[\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}} - 1\}$$

The FOC for an individual $L(i)$ is

$$p(i) = \lambda L(i)^{\varphi-1} \quad (3.31)$$

Explanation:

According to FOC

$$\begin{aligned} \frac{\delta \mathcal{L}}{\delta L(i)} &= \frac{\delta}{\delta L(i)} (\int_{i=0}^A p(i)L(i)di - \lambda \{[\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}} - 1\}) = 0 \\ &\Rightarrow \frac{\delta}{\delta L(i)} \int_{i=0}^A p(i)L(i)di - \frac{\delta}{\delta L(i)} \lambda \{[\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}} - 1\} = 0 \\ &\Rightarrow p(i) - \lambda \frac{1}{\varphi} [\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}-1} \varphi L(i)^{\varphi-1} = 0 \\ &\Rightarrow p(i) - \lambda L(i)^{\varphi-1} = 0 \quad [\text{since } [\int_{i=0}^A L(i)^\varphi di]^{\frac{1}{\varphi}} = 1] \\ &\Rightarrow p(i) = \lambda L(i)^{\varphi-1} \end{aligned}$$

Equation (3.31) implies $L(i)^{\varphi-1} = \frac{p(i)}{\lambda}$ which in turn implies

$$\begin{aligned} L(i) &= [\frac{p(i)}{\lambda}]^{\frac{1}{\varphi-1}} \\ &= [\frac{\lambda}{p(i)}]^{\frac{1}{1-\varphi}} \end{aligned} \quad (3.32)$$

- Equation (3.32) shows that
 - the holder of the patent on an idea faces a downward-sloping demand curve
 - $L(i)$ is a smoothly **decreasing** function of $p(i)$
 - When φ is **closer to 1**
 - the marginal product of an input declines more slowly as the quantity of the input rises.
 - As a result, the inputs are closer substitutes, and so
 - the **elasticity of demand for each input is greater**
- Because
 - firms **producing final output** face **constant costs** for each input and
 - **the production function** exhibits **constant returns**,
 - **marginal cost equals average cost**
 - As a result, **these firms earn zero profits**.

The Rest of the Model

The remainder of the model involves **four sets of assumptions**.

i) The first set of assumption:

The first set concern **economic aggregates**.

- **Population is fixed** and equal to $L > 0$
- We let $L_A(t)$ denote the number of workers **engaged in R&D** at time t
- And $L_Y(t) = \int_{i=0}^{A(t)} L(i)^\varphi di$ is the total number of workers **producing inputs** at time t
- then **equilibrium** in the labor market at t requires

$$L_A(t) + L_Y(t) = \bar{L} \quad (3.33)$$

- The **production function** for **new ideas** is
 - **linear in L_A** (the number of workers employed in R&D)
 - **proportional to A** (the existing stock of knowledge)

$$\dot{A}(t) = B L_A(t) A(t) \quad B > 0 \quad (3.34)$$

- the initial level of A , $A(0)$, is assumed to be strictly positive.

- These assumptions are chosen to give the model the **aggregate dynamics** of a **linear growth model**.

ii) The second set of assumption:

The second group of assumptions concern **the microeconomics of household behavior**.

- The behavior is **similar** to that of **RCK economy**
 - Individuals are infinitely lived
 - maximize a conventional utility function
 - Individuals' discount rate is ρ

- their instantaneous utility function is logarithmic.
- the representative individual's **lifetime utility** is

$$U = \int_{t=0}^{\infty} e^{-\rho t} \ln C(t) dt \quad \rho > 0 \quad (3.35)$$

- As in the RCK model, the individual's **budget constraint** is

$$\int_{t=0}^{\infty} e^{-r(t)} c(t) dt \leq X(0) + \int_{t=0}^{\infty} e^{-r(t)} w(t) dt \quad (3.36)$$

Where, r is the interest rate

$X(0)$ is initial wealth per person, and

$w(t)$ is the wage at t

iii) The third set of assumption:

The third set of assumptions concern **the microeconomics of R&D**.

- There is **free entry into idea creation**
 - anyone can hire $\frac{1}{BA(t)}$ units of labor at the prevailing wage $w(t)$
 - and produce a new idea (see [3.34])
 - the cost of creating that idea will be $\frac{w(t)}{BA(t)}$
- The creator of an idea is granted **permanent patent rights** to the use of the idea
- The free-entry condition in R&D requires that
 - the **present value of the profits** earned from selling the **input embodying an idea** equals the **cost of creating it**.
- If idea i is created at time t
 - and let $\pi(i, \tau)$ denote the **profits** earned by the creator of the idea at time τ
 - Then this condition is

$$\int_{\tau=t}^{\infty} e^{-r(\tau-t)} \pi(i, \tau) d\tau = \frac{w(t)}{BA(t)} \quad (3.37)$$

iv) The fourth set of assumption:

The final assumptions of the model concern **general equilibrium**

- **First**, the assumption that the labor market is competitive
 - implies that the wage paid in R&D and the wages paid by all input producers are equal.
- **Second**, the only asset in the economy is the patents.
- **Finally**, the only use of the output good is for consumption
 - Equilibrium in the goods market at time t requires

$$C(t) \bar{L} = Y(t) \quad (3.38)$$

Solving the Model

The problem of a patent holder

Price of input

- The first step in solving the model is to consider **the problem of a patent holder**
 - **choosing the price** to charge for his or her input at a point in time.
- A standard result from microeconomics is that
 - the **profit-maximizing price** of a monopolist is $\frac{\eta}{\eta - 1}$ **times marginal cost**
 - where η is the **elasticity of demand**
- In our case, **the elasticity of demand** is constant and equal to $\frac{1}{1 - \varphi}$
- And the **marginal cost** at time t is $w(t)$
- Each monopolist therefore charges (price)

$$\frac{\frac{1}{\frac{1}{1 - \varphi} - 1}}{\frac{1}{1 - \varphi} - 1} w(t), \text{ or } \frac{w(t)}{\varphi}$$

Explanation:

$$\begin{aligned} \frac{\frac{1}{\frac{1}{1 - \varphi} - 1}}{\frac{1}{1 - \varphi} - 1} w(t) &= \frac{\frac{1}{1 - \varphi}}{\frac{1 - (1 - \varphi)}{1 - \varphi}} w(t) \\ &= \frac{\frac{1}{1 - \varphi}}{\frac{\varphi}{1 - \varphi}} w(t) \\ &= \frac{1}{1 - \varphi} \cdot \frac{1 - \varphi}{\varphi} \cdot w(t) \\ &= \frac{w(t)}{\varphi} \end{aligned}$$

Patent-holder's profits

Knowing the price each monopolist charges

- allows us to determine his or her **profits** at a point in time.

- Given our assumption that
 - L_A is constant and
 - the requirement that $L_A(t) + L_Y(t) = \bar{L}$
 - The quantity of each input used at time t is

$$\frac{\bar{L} - L_A}{A(t)}$$

- **Each patent-holder's profits** are thus

$$\begin{aligned}\pi(t) &= \frac{\bar{L} - L_A}{A(t)} \left[\frac{w(t)}{\varphi} - w(t) \right] \\ &= \frac{1 - \varphi}{\varphi} \frac{\bar{L} - L_A}{A(t)} w(t)\end{aligned}\tag{3.39}$$

Explanation:

Patent-holder's profit = Patent-holder's revenue — Patent-holder's cost

$$\begin{aligned}\pi(t) &= \frac{\bar{L} - L_A}{A(t)} \frac{w(t)}{\varphi} - \frac{\bar{L} - L_A}{A(t)} w(t) \\ &= \frac{\bar{L} - L_A}{A(t)} \left[\frac{w(t)}{\varphi} - w(t) \right] \\ &= \frac{\bar{L} - L_A}{A(t)} \left[\frac{1}{\varphi} - 1 \right] w(t) \\ &= \frac{1 - \varphi}{\varphi} \frac{\bar{L} - L_A}{A(t)} w(t)\end{aligned}$$

Present value of profits

- To determine the present value of profits from an invention, and hence the incentive to innovate,
- we need to determine
 - **the economy's growth rate** and
 - **the interest rate**

Economy's growth rate

According to Equation (3.34) $\dot{A}(t) = BL_A(t) A(t)$

- This implies that
 - if L_A is constant,
 - Growth rate of A is, $\frac{\dot{A}(t)}{A(t)} = BL_A$

According to Equation (3.29) $Y(t) = A(t)^{\frac{1-\varphi}{\varphi}} L_Y(t)$

- Since $L_Y(t)$ is constant
 - the growth rate of Y is $\frac{1-\varphi}{\varphi}$ times the growth rate of A (which is, BL_A)
 - Growth rate of Y is, $\frac{\dot{Y}(t)}{Y(t)} = \frac{1-\varphi}{\varphi} BL_A$

The interest rate

- ❖ Both **consumption** and the **wage grow at the same rate** as **output**.
- ❖ And the **growth rate of the wage equals** the **growth rate of output**
- ❖ Thus, **Growth rate of output = Growth rate of consumption**

❖ Thus,
$$\frac{\dot{Y}(t)}{Y(t)} = \frac{\dot{C}(t)}{C(t)} = \frac{\dot{w}(t)}{w(t)} \frac{1-\varphi}{\varphi} BL_A$$

- Once we know the growth rate of consumption, finding the **real interest rate** is straightforward.
- Euler Equation of the RCK model can be written as:

$$\frac{\dot{C}(t)}{C(t)} = \frac{r(t) - \rho}{\theta}$$

➤ With logarithmic utility, θ is 1

- Thus equilibrium requires

$$\begin{aligned} r(t) &= \rho + \frac{\dot{C}(t)}{C(t)} \\ &= \rho + \frac{1-\varphi}{\varphi} BL_A \end{aligned} \tag{3.40}$$

- Thus if L_A is **constant**, the **real interest rate** is **constant**

Explanation:

$$\frac{\dot{C}(t)}{C(t)} = \frac{r(t) - \rho}{\theta}$$

$$r(t) - \rho = \frac{\dot{C}(t)}{C(t)} \quad (\text{when } \theta = 1)$$

$$r(t) = \rho + \frac{\dot{C}(t)}{C(t)}$$

$$r(t) = \rho + \frac{1-\varphi}{\varphi} BL_A \quad (\text{Since, Growth rate of output} = \text{Growth rate of consumption})$$

Present value of profits

- According to Equation (3.39) profits at t are $\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{A(t)} w(t)$
- Growth rate of profit = Growth rate of w - Growth rate of A

$$\begin{aligned} &= \frac{1-\varphi}{\varphi} BL_A - BL_A \\ &= \left[\frac{1-\varphi}{\varphi} - 1 \right] BL_A \\ &= \left[\frac{1-\varphi-\varphi}{\varphi} \right] BL_A \\ &= \frac{1-2\varphi}{\varphi} BL_A \end{aligned}$$

- Profits are discounted at the interest rate r . (i.e., at rate $\rho + \frac{1-\varphi}{\varphi} BL_A$)
- The **present value of the profits** earned from the discovery of a new idea at time t is therefore

$$\begin{aligned} \text{PV of } \pi(t) &= \frac{\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{A(t)} w(t)}{\rho + \frac{1-\varphi}{\varphi} BL_A - \frac{1-2\varphi}{\varphi} BL_A} \\ &= \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{\rho + BL_A} \frac{w(t)}{A(t)} \end{aligned} \quad (3.41)$$

Explanation:

Present value of profit at time $t = \frac{\text{profit at time } t}{\text{discount rate} - \text{growth rate of profit}}$

$$\begin{aligned} \text{PV of } \pi(t) &= \frac{\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{A(t)} w(t)}{\rho + \frac{1-\varphi}{\varphi} BL_A - \frac{1-2\varphi}{\varphi} BL_A} \\ &= \frac{\frac{1-\varphi}{\varphi} \bar{L}-L_A \frac{w(t)}{A(t)}}{\rho + BL_A \left[\frac{1-\varphi}{\varphi} - \frac{1-2\varphi}{\varphi} \right]} \\ &= \frac{\frac{1-\varphi}{\varphi} \bar{L}-L_A \frac{w(t)}{A(t)}}{\rho + BL_A \left[\frac{1-\varphi-1+2\varphi}{\varphi} \right]} \\ &= \frac{\frac{1-\varphi}{\varphi} \bar{L}-L_A \frac{w(t)}{A(t)}}{\rho + BL_A \left[\frac{\varphi}{\varphi} \right]} \\ &= \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{\rho + BL_A} \frac{w(t)}{A(t)} \end{aligned}$$

Equilibrium value of L_A

- If the amount of **R &D** is strictly **positive**
 - **the present value of profits** from an invention **must equal the costs of the invention**
- Since one worker can produce $BA(t)$ ideas per unit time
 - the cost of an invention is $\frac{w(t)}{BA(t)}$
- The equilibrium condition is therefore

$$\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{\rho + BL_A} \frac{w(t)}{A(t)} = \frac{w(t)}{BA(t)} \quad (3.42)$$

Solving this equation for L_A yields

$$L_A = (1-\varphi) \bar{L} - \frac{\varphi\rho}{B} \quad (3.43)$$

Explanation:

$$\begin{aligned}\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{\rho+BL_A} \frac{w(t)}{A(t)} &= \frac{w(t)}{BA(t)} \\ \Rightarrow \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{\rho+BL_A} &= \frac{w(t)}{BA(t)} \cdot \frac{A(t)}{w(t)} \\ \Rightarrow \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{B(\frac{\rho}{B}+L_A)} &= \frac{1}{B} \\ \Rightarrow \frac{1}{B} \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{(\frac{\rho}{B}+L_A)} &= \frac{1}{B} \\ \Rightarrow \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{(\frac{\rho}{B}+L_A)} &= 1 \\ \Rightarrow (1-\varphi)(\bar{L}-L_A) &= \varphi(\frac{\rho}{B}+L_A) \\ \Rightarrow (1-\varphi)\bar{L} - (1-\varphi)L_A &= \frac{\varphi\rho}{B} + \varphi L_A \\ \Rightarrow (1-\varphi)\bar{L} - \frac{\varphi\rho}{B} &= \varphi L_A + (1-\varphi)L_A \\ \Rightarrow (1-\varphi)\bar{L} - \frac{\varphi\rho}{B} &= L_A(\varphi + 1 - \varphi) \\ \Rightarrow L_A &= (1-\varphi)\bar{L} - \frac{\varphi\rho}{B}\end{aligned}$$

- However, the amount of R&D need not be strictly positive.
- In particular, when (3.43) implies $L_A < 0$
 - R & D will be 0
- Thus we can modify (3.43) to

$$L_A = \max \left\{ (1-\varphi)\bar{L} - \frac{\varphi\rho}{B}, 0 \right\} \quad (3.44)$$

Finally, since the growth rate of output is $\frac{1-\varphi}{\varphi} BL_A$, we have

$$\frac{\dot{Y}(t)}{Y(t)} = \max \left\{ \frac{(1-\varphi)^2}{\varphi} B\bar{L} - (1-\varphi)\rho, 0 \right\} \quad (3.45)$$

- Thus we have succeeded in describing **how long-run grow this determined**
- Since none of the terms on the right-hand side of (3.43) are time-varying
 - the equilibrium value of L_A is **constant**

Explanation:

$$\begin{aligned}\frac{\dot{Y}(t)}{Y(t)} &= \frac{1-\varphi}{\varphi} B L_A \\ &= \frac{1-\varphi}{\varphi} B \cdot \max \left\{ (1-\varphi) \bar{L} - \frac{\varphi \rho}{B}, 0 \right\} \\ &= \max \left\{ \frac{(1-\varphi)^2}{\varphi} B \bar{L} - \frac{1-\varphi}{\varphi} B \frac{\varphi \rho}{B}, 0 \right\} \\ &= \max \left\{ \frac{(1-\varphi)^2}{\varphi} B \bar{L} - (1-\varphi) \rho, 0 \right\}\end{aligned}$$

Implications

The model has **two major** sets of implications

❖ *The first set of implication:*

The first concern the **determinants of long-run growth**.

- **Four parameters** affect the **economy's growth rate**

First, when individuals are less patient

- that is, **when ρ is higher**
 - fewer workers engage in R&D (equation [3.44])
 - **growth is lower** (equation [3.45])

Second, an **increase** in substitutability among inputs (φ) also **reduces growth**

There are **two reasons**

- fewer** workers engage in R&D (equation [3.44])
- although a given amount of R&D translates into the same growth rate of A (equation [3.34]), **a given growth rate of A** translates into **slower output growth** (equation [3.29])

Third, an **increase in productivity** in the R&D sector (**B**) **increases growth**.

There are again **two effects** at work.

- a rise in B raises growth for a given number of workers engaged in R&D.
- increased productivity in R&D draws more workers into that sector.

Finally, an **increase** in the size of the population (**L**) **raises** long-run growth.

There are **two effects**

- i) growth increases for a given fraction of workers engaged in R&D, and the fraction of workers engaged in R&D
- ii) an increase in the size of the economy expands the market an inventor can reach, and so increases the returns to R&D

❖ **The second set of implication:**

- The model's second major set of implications concern
 - **the gap** between **equilibrium** and **optimal growth**
- Since the economy is not perfectly competitive, **there is no reason to expect the decentralized equilibrium to be socially optimal**.
- let us look for the constant level of **L_A** that yields the **highest level of lifetime utility** for the representative individual.
- Because all output is consumed
 - the representative individual's **consumption** is $\frac{1}{L}$ times output
 - Equation (3.29) for output therefore implies that the representative individual's **consumption at time 0** is

$$C(0) = \frac{(\bar{L} - L_A) A(0)^{\frac{1-\varphi}{\varphi}}}{\bar{L}} \quad (3.46)$$

Explanation:

The representative individual's consumption $= \frac{1}{L}$ times output

Therefore, $C(0) = \frac{1}{L} \cdot Y(0)$

$$= \frac{A(0)^{\frac{1-\varphi}{\varphi}} L_Y}{\bar{L}} \quad [\text{from (3.29)}]$$

$$= \frac{A(0)^{\frac{1-\varphi}{\varphi}} (\bar{L} - L_A)}{\bar{L}} \quad [\text{from (3.33)}]$$

$$= \frac{(\bar{L} - L_A) A(0)^{\frac{1-\varphi}{\varphi}}}{\bar{L}}$$

- The representative individual's lifetime utility is therefore

$$U = \int_0^{\infty} e^{-\rho t} \ln \left[\frac{(\bar{L} - L_A)}{\bar{L}} A(0)^{\frac{1-\varphi}{\varphi}} e^{\left[\frac{1-\varphi}{\varphi}\right] B L_A t} \right] dt \quad (3.47)$$

Explanation:

From (3.35)

$$U = \int_{t=0}^{\infty} e^{-\rho t} \ln C(t) dt$$

$$\text{or, } U = \int_{t=0}^{\infty} e^{-\rho t} \ln [C(0) e^{\frac{\dot{C}(t)}{C(t)} t}] dt$$

$$U = \int_{t=0}^{\infty} e^{-\rho t} \ln \left[\frac{(\bar{L} - L_A) A(0)^{\frac{1-\varphi}{\varphi}}}{\bar{L}} e^{\frac{\dot{C}(t)}{C(t)} t} \right] dt \quad [\text{From (3.46)}]$$

$$U = \int_0^{\infty} e^{-\rho t} \ln \left[\frac{(\bar{L} - L_A)}{\bar{L}} A(0)^{\frac{1-\varphi}{\varphi}} e^{\left[\frac{1-\varphi}{\varphi}\right] B L_A t} \right] dt \quad \left[\text{since, } \frac{\dot{C}(t)}{C(t)} = \frac{1-\varphi}{\varphi} B L_A \right]$$

- One can show that the solution to this integral is

$$U = \frac{1}{\rho} \left(\ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1-\varphi}{\varphi} \ln A(0) + \frac{1-\varphi}{\varphi} \frac{B L_A}{\rho} \right) \quad (3.48)$$

Explanation:

From (3.47)

$$U = \int_0^{\infty} e^{-\rho t} \ln \left[\left(\frac{\bar{L} - L_A}{\bar{L}} \right) A(0)^{\frac{1-\varphi}{\varphi}} e^{\left[\frac{1-\varphi}{\varphi} \right] B L_A t} \right] dt$$

$$U = \int_0^{\infty} e^{-\rho t} \ln \left(\frac{\bar{L} - L_A}{\bar{L}} \right) dt + \int_0^{\infty} e^{-\rho t} \frac{1-\varphi}{\varphi} \ln A(0) dt + \int_0^{\infty} e^{-\rho t} \frac{1-\varphi}{\varphi} B L_A t dt$$

$$U = \ln \frac{\bar{L} - L_A}{\bar{L}} \int_0^{\infty} e^{-\rho t} dt + \frac{1-\varphi}{\varphi} \ln A(0) \int_0^{\infty} e^{-\rho t} dt + \frac{1-\varphi}{\varphi} B L_A \int_0^{\infty} e^{-\rho t} \cdot t dt$$

$$U = \ln \frac{\bar{L} - L_A}{\bar{L}} \frac{1}{\rho} + \frac{1-\varphi}{\varphi} \ln A(0) \frac{1}{\rho} + \frac{1-\varphi}{\varphi} B L_A \int_0^{\infty} t \cdot e^{-\rho t} dt \quad \left[\text{since, } \int_0^{\infty} e^{-xt} = \frac{1}{x} \right]$$

$$U = \frac{1}{\rho} \ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1}{\rho} \frac{1-\varphi}{\varphi} \ln A(0) + \frac{1-\varphi}{\varphi} B L_A \int_0^{\infty} t \cdot e^{-\rho t} dt \dots\dots\dots (i)$$

We know $\int u dv = uv - \int v du$

Here we have, $\int_0^{\infty} t \cdot e^{-\rho t} dt$

Where, $u = t$ and $dv = e^{-\rho t}$

Therefore, $\frac{du}{dt} = 1$ and $\int dv = \int e^{-\rho t} dt$

$du = dt$ and $v = -\frac{e^{-\rho t}}{\rho}$ [since, $\int e^{mx} = \frac{e^{mx}}{m}$]

Thus, $\int t \cdot e^{-\rho t} = \left[t \cdot \frac{-1}{\rho} (e^{-\rho t}) \right] - \int \frac{-1}{\rho} e^{-\rho t} dt$

$$\begin{aligned} \text{Therefore, } \int_0^{\infty} t \cdot e^{-\rho t} dt &= \left[t \cdot \frac{-1}{\rho} (e^{-\rho t}) \right]_0^{\infty} - \int_0^{\infty} \frac{-1}{\rho} e^{-\rho t} dt \\ &= \left[\infty \cdot \frac{-1}{\rho} (e^{-\rho \infty}) - 0 \cdot \frac{-1}{\rho} (e^{-\rho \cdot 0}) \right] - \int_0^{\infty} \frac{-1}{\rho} e^{-\rho t} dt \\ &= [0 - 0] - \int_0^{\infty} \frac{-1}{\rho} e^{-\rho t} dt \quad \left[\text{since } e^{-\infty} = 0 \right] \\ &= \int_0^{\infty} \frac{1}{\rho} e^{-\rho t} dt \\ &= \frac{1}{\rho} \int_0^{\infty} e^{-\rho t} dt \\ &= \frac{1}{\rho} \cdot \frac{1}{\rho} \quad \left[\text{since, } \int_0^{\infty} e^{-xt} = \frac{1}{x} \right] \\ \int_0^{\infty} t \cdot e^{-\rho t} dt &= \frac{1}{\rho^2} \dots\dots\dots (ii) \end{aligned}$$

Now, by substituting the value of (ii) into (i) we obtain

$$U = \frac{1}{\rho} \ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1}{\rho} \frac{1-\varphi}{\varphi} \ln A(0) + \frac{1-\varphi}{\varphi} B L_A \frac{1}{\rho^2}$$

$$U = \frac{1}{\rho} \left(\ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1-\varphi}{\varphi} \ln A(0) + \frac{1-\varphi}{\varphi} \frac{B L_A}{\rho} \right)$$

- Maximizing this expression with respect to L_A shows that the socially optimal level of L_A is given by

$$L_A^{OPT} = \max \left\{ \bar{L} - \frac{\varphi}{1-\varphi} \frac{\rho}{B}, 0 \right\} \quad (3.49)$$

Explanation:

According to FOC:

$$\begin{aligned} \frac{dU}{dL_A} &= \frac{d}{dL_A} \left[\frac{1}{\rho} \left(\ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1-\varphi}{\varphi} \ln A(0) + \frac{1-\varphi}{\varphi} \frac{BL_A}{\rho} \right) \right] = 0 \\ &\Rightarrow \frac{d}{dL_A} \left(\ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1-\varphi}{\varphi} \ln A(0) + \frac{1-\varphi}{\varphi} \frac{BL_A}{\rho} \right) = 0 \\ &\Rightarrow \frac{d}{dL_A} \left[\ln (\bar{L} - L_A) - \ln \bar{L} \right] + \frac{d}{dL_A} \frac{1-\varphi}{\varphi} \ln A(0) + \frac{d}{dL_A} \frac{1-\varphi}{\varphi} \frac{BL_A}{\rho} = 0 \\ &\Rightarrow \frac{1}{\bar{L} - L_A} (-1) + 0 + \frac{1-\varphi}{\varphi} \frac{B}{\rho} = 0 \\ &\Rightarrow \frac{1}{\bar{L} - L_A} = \frac{1-\varphi}{\varphi} \frac{B}{\rho} \\ &\Rightarrow \bar{L} - L_A = \frac{\varphi}{1-\varphi} \frac{\rho}{B} \\ &\Rightarrow L_A = \bar{L} - \frac{\varphi}{1-\varphi} \frac{\rho}{B} \end{aligned}$$

Therefore, the socially optimal level of L_A is

$$L_A^{OPT} = \max \left\{ \bar{L} - \frac{\varphi}{1-\varphi} \frac{\rho}{B}, 0 \right\}$$

- Comparing this expression with equation (3.44)

$$L_A^{EQ} = (1 - \varphi) L_A^{OPT} \quad (3.50)$$

Explanation:

From (3.44), equilibrium level of L_A

$$\begin{aligned} L_A^{EQ} &= \left\{ (1-\varphi) \bar{L} - \frac{\varphi \rho}{B}, 0 \right\} \\ L_A^{EQ} &= (1-\varphi) \left[\bar{L} - \frac{\varphi}{(1-\varphi)} \frac{\rho}{B} \right] \\ L_A^{EQ} &= (1-\varphi) L_A^{OPT} \end{aligned}$$

- If $\varphi > \frac{1}{2}$
 - the profits of existing patent-holders are reduced by an increase in A
- If $\varphi < \frac{1}{2}$
 - the profits of existing patent-holders are raised by an increase in A
- The equilibrium number of workers engaged in R&D is
 - always less than the optimal number
- Thus growth is always in efficiently low
- Moreover, the proportional gap between the equilibrium and optimal numbers (and hence between equilibrium and optimal growth)
 - depends only on a single parameter, φ
 - The smaller the degree of differentiation among inputs embodying different ideas (that is, the greater is φ), the greater the gap

Extensions

First, Introducing physical capital does not change the model's central messages.

- In Romer's model, where physical capital is not an input into R&D,
 - policies that increase physical-capital investment have only level effects, not growth effects.
- In variants where capital enters the production function for ideas, such policies generally have growth effects

Second,

Extending the model into a nonlinear version changes its implication.

- We have been considering correspond to a linear growth model.
- Jones (1995a) extends the Romer model to the case
 - where the exponent on A in the production function for ideas is less than 1.
 - This creates transition dynamics
 - More importantly, it changes the model's messages concerning the determinants of long-run growth.
 - The macroeconomics of Jones's model correspond to those of a semi-endogenous growth model.
 - As a result, long-run growth depends only on the rate of population growth.

Third,

If technology improves existing inputs, the model may identify other factors that affect long-run economic performance.

- In Romer's model, technological progress takes the form of expansion of the number of inputs into production.
- An alternative is that it takes the form of improvements in existing inputs.

- This leads to the “**quality ladder**” models of Grossman and Helpman (1991a) and Aghion and Howitt (1992).
- Quality-ladder models do not produce sharply different answers than expanding-variety models concerning the long-run growth and level of income.
 - But they identify additional microeconomic determinants of incentives for innovation, and so
 - show **other factors that affect long-run economic performance**.